

TASK – 3

FORENSIC REPORT – MULTI-STAGE APT ATTACK

1. Introduction

This report presents the forensic investigation of a multi-stage advanced persistent threat (APT) attack using open-source SOC tools. The attack chain includes initial access, persistence, privilege escalation, exfiltration and lateral movement.

2. Scope of Investigation

- Generate evidence for initial access, persistence, privilege escalation, exfiltration and lateral movement.
- Detect suspicious activity using Security Onion and Wazuh.
- Use Kibana for hunting indicators of compromise.
- Create an incident case in TheHive and use cortex for the IOC workflow.
- Document IOC sharing using MISP.
- Reconstruct attack timeline.
- Document forensic findings.

3. Tools Used

1. **Security Onion** – provided an SOC platform for the project.
2. **Zeek** – network traffic analysis.
3. **Suricata** – generated alerts during attack.
4. **Wazuh Cloud** – host-based monitoring system.
5. **TheHive & Cortex** – used for incident case management.
6. **Docker** – to deploy security tools.
7. **MISP** – to share extracted IOCs.
8. **Oracle VM VirtualBox** – runs virtual machine on it.

4. Evidence Sources

1. Host – auth logs, bash history, cron entries, suspicious scripts.
2. Network – DNS queries, HTTP traffic, data transfer attempts.
3. Tools – Security onion alerts, Wazuh dashboards, Suricata events, Zeek metadata.

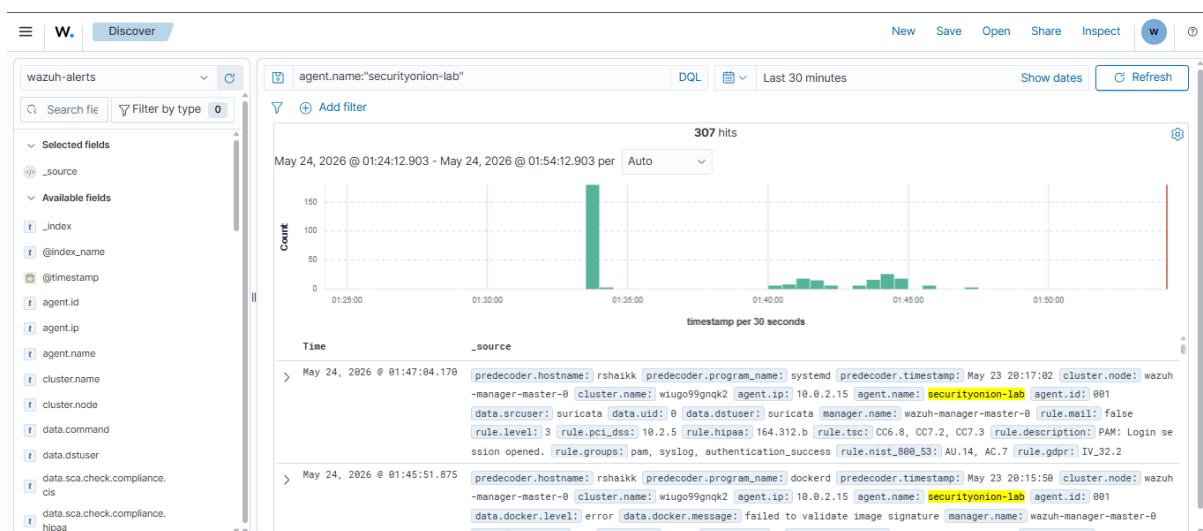
5. Reconstructed Attack timeline

STEP 1 – Initial access.

- Security Onion -> Alerts showed the indication of suspicious outbound HTTP traffic using Suricata, and IDS system.
- This indicates possible malicious payload download.

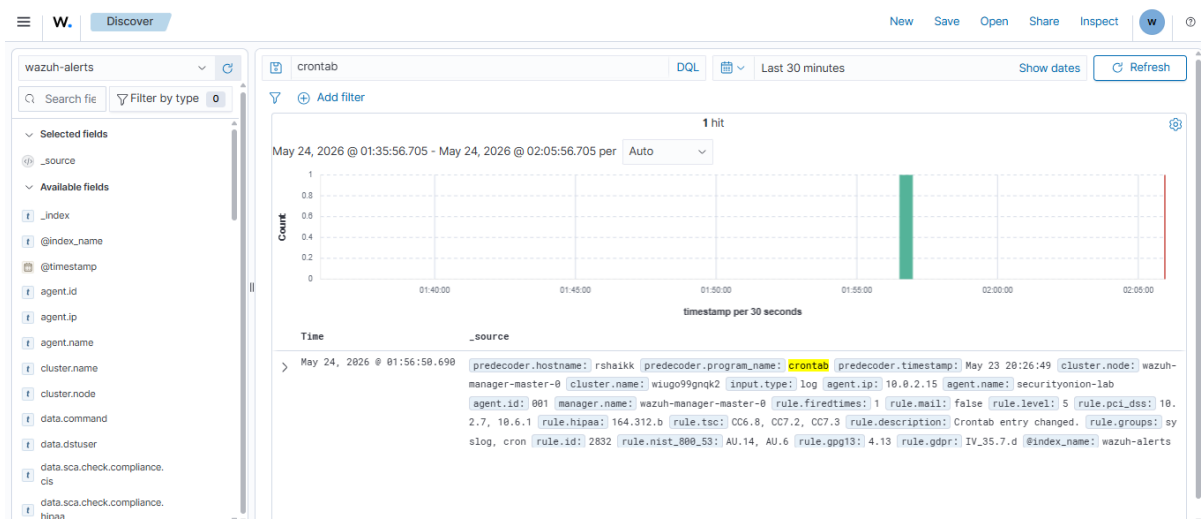
Timestamp	event.dataset	source.ip	source.port	destination.ip	destination.port	rule.name
2026-05-23 07:49:44.253 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:44:19.277 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:38:54.251 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:33:29.275 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:28:04.300 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:22:39.272 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:17:14.296 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:11:49.320 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:06:24.343 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S
2026-05-23 07:00:59.315 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621	ET INFO S

Wazuh showed events on searching for **agent.name:"securityonion-lab"**.



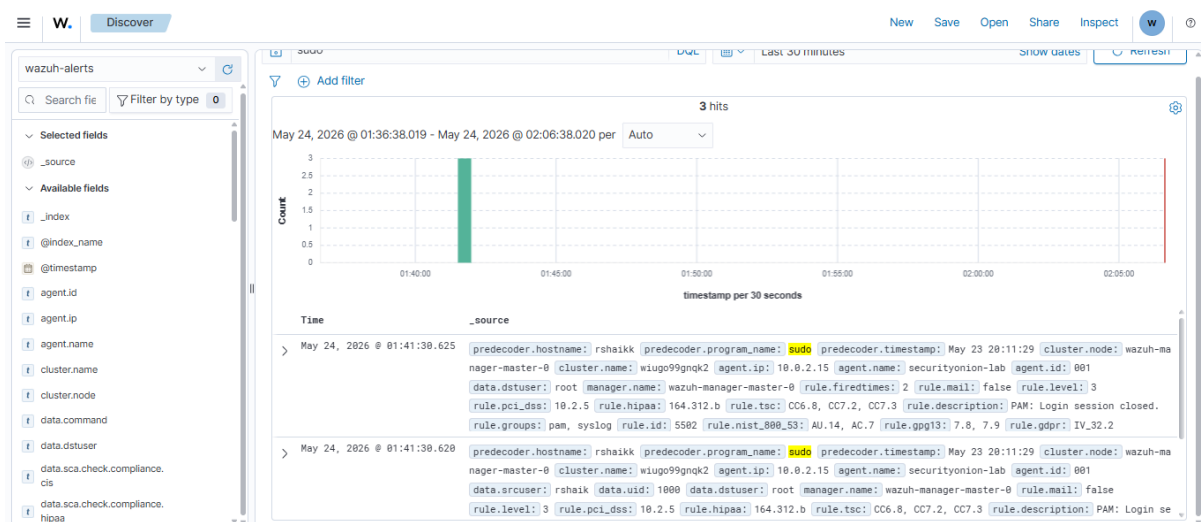
STEP 2 - Persistence

- Wazuh successfully detected cron-based persistence through creation of scheduled tasks indicating suspicious recurring execution of the malicious script.
- This indicates the attacker persistence mechanism on adding the filter **crontab** in wazuh.



STEP 3 – Privilege escalation

Wazuh detection of privilege escalation activity through **sudo** command execution thus indicating suspicious privilege activities by the attacker.



STEP 4 – Lateral movement

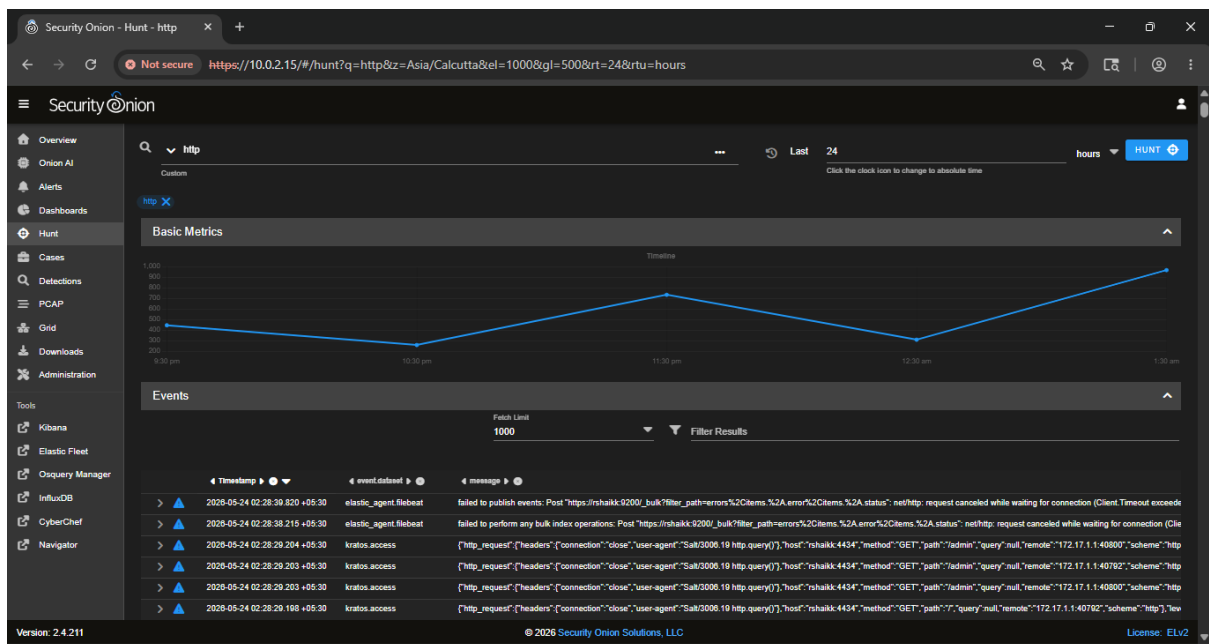
Using Security Onion Alerts and Wazuh, we could detect the SSH failed login attempts by the attacker indicating lateral movement in the victim system through unauthorized remote access attempts.

Count	rule_name	event.module	event.severity_label	rule.uuid
4	Security Onion - Grid Node Login Failure (SSH)	sigma	high	923421c7-9b1e-45
ALERT DETAILS				
@timestamp	2026-05-23T17:50:07.000Z			
event.agent_id_status	missing			
event.dataset	sigma.alert			
event.ingested	2026-05-23T17:50:07Z			
event.module	sigma			
event.severity	4			
event.severity_label	high			
event_data.@timestamp	2026-05-23T17:48:12Z			
event_data.id	lzn0VZ4BY73f36_ysNFY			
event_data.index	.ds-logs-system.auth-default-2026.05.18-000001			
event_data.agent.ephemeral_id	2f1aa776-cb08-4219-b6d8-a03887d180f4			
event_data.agent.id	ca89553d-2489-446d-8c7a-9696c4a316b0			
event_data.agent.name	rshaikk			

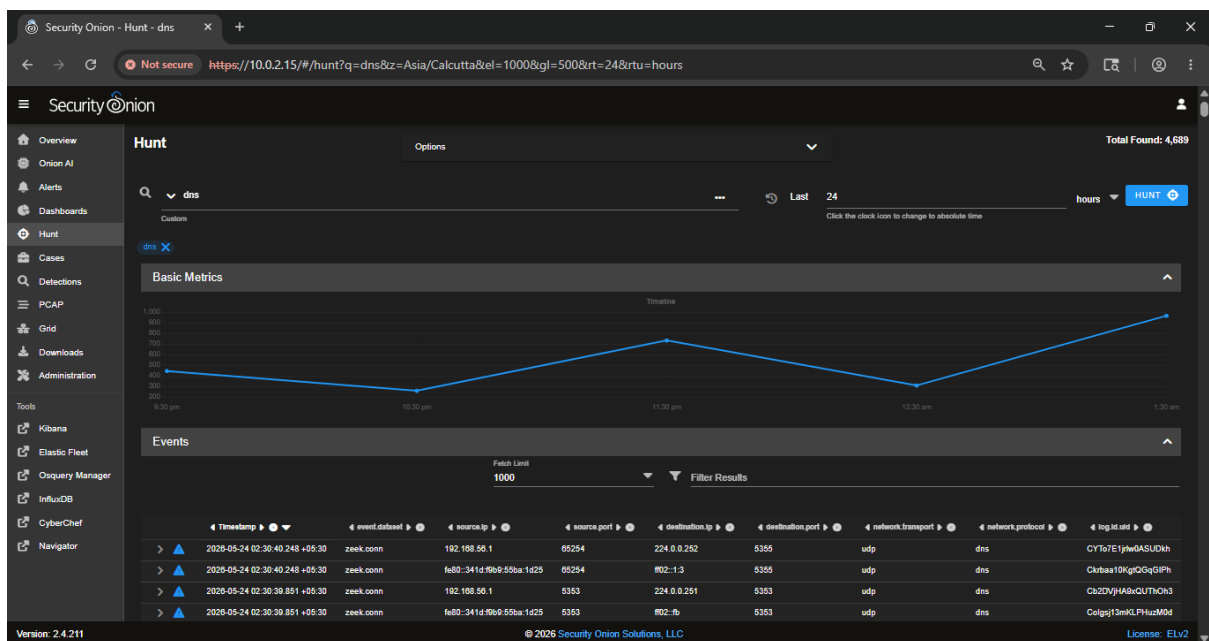


STEP 5 – Exfiltration

Security Onion showed outbound **HTTP** communication with GET method indicating possible exfiltration attempts.



Also, unusual **DNS** traffic indicating possible DNS Tunnelling activities.



6. IOC Extraction

Security Onion -> Hunt, this is Kibana-backed hunting in Security Onion.

HUNT 1 – Initial Access IOC

Filter used: http.

Timestamp	event.dataset	source.ip	source.port	destination.ip	destination.port
2026-05-24 14:16:23.483 +05:30	suricata.alert	192.168.56.1	57621	192.168.56.255	57621
@timestamp	2026-05-24T08:46:23.488Z				
data_stream.dataset	suricata				
data_stream.namespace	so				
data_stream.type	logs				
destination.ip	192.168.56.255				
destination.port	57621				
dns.query_name	[53 70 6f 74 55 64 70 30]				
ecs.version	8.0.0				
elastic_agent.id	ca89553d-2489-446d-8c7a-9696c4a316b0				
elastic_agent.snapshot	false				
elastic_agent.version	9.0.8				
event.agent_id_status	missing				
event.category	network				
event.dataset	suricata.alert				
eventIngested	2026-05-24T08:47:44Z				

HUNT 2 – Persistence IOC

Filter used: cron.

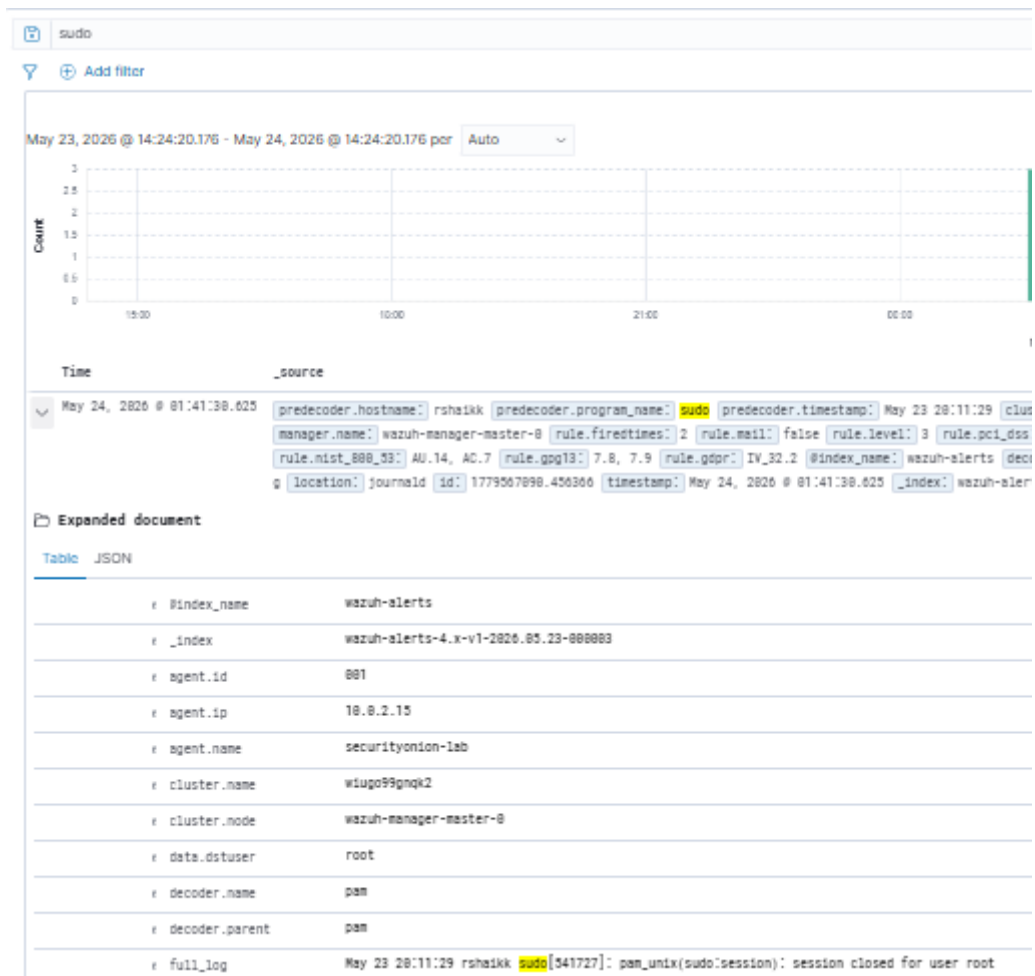


IOC Extracted

- agent: securityonion-lab
- process: crontab
- MITRE: T1053 Scheduled Task / cron

HUNT 3 – Privilege Escalation IOC

Filter used: sudo

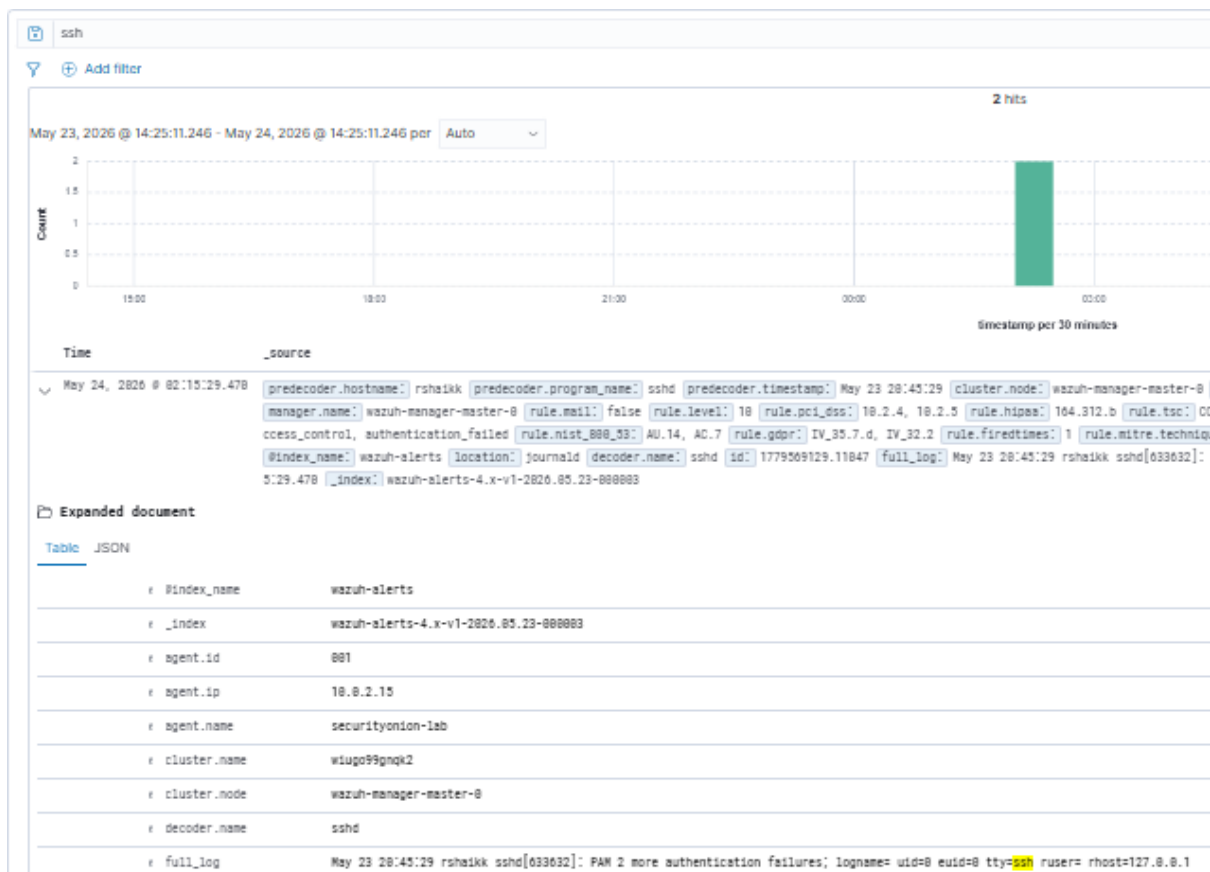


IOC Extracted

- agent IP: 10.0.2.15
- command used: sudo
- rule description: PAM: Login session closed
- user type: root
- user account: rshaikk
- MITRE: T1548 Abuse Elevation Control Mechanism

HUNT 4 – Lateral Movement IOC

Filter used: ssh



IOC Extracted

- program: ssh
- authentication status: failed login
- protocol: SSH
- MITRE: T1021 Remote Services

HUNT 5 – Exfiltration IOC

Filter used: dns

network.protocol	dns
network.transport	udp
observer.name	rshaikk
pipeline	conn
server.ip	ff02::1:3
server.ip_bytes	0
server.packets	0
server.port	5355
source.ip	fe80::341d:f9b9:55ba:1d25
source.port	49694
tags	["conn"]
soc_id	wzuxVp4BY73f36_yK5bd
soc_score	2
soc_type	
soc_timestamp	2026-05-23T21:13:20.289Z
soc_source	rshaikk.ds-logs-zeek-so-2026.05.18-000001
> 2026-05-24 02:43:20.289 +05:30 zeek.conn 192.168.56.1	

IOC Extracted

protocols: HTTP, DNS, UDP

source IP:192.168.56.1

HTTPS destination: example.com

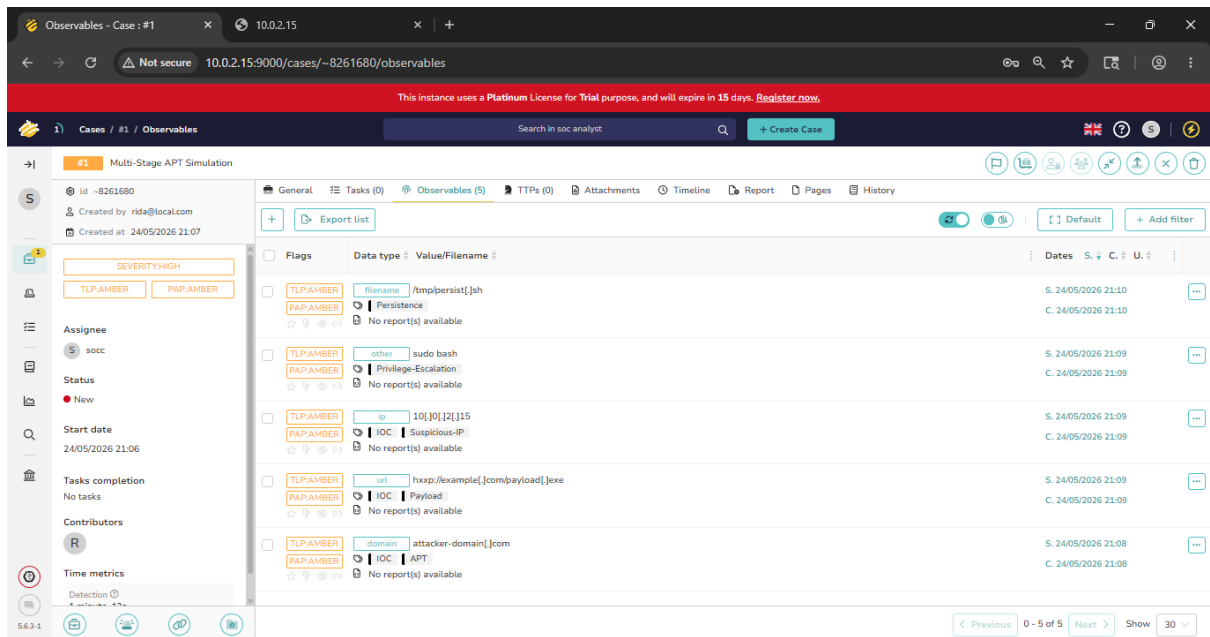
MITRE: T1048 — Exfiltration Over Alternative Protocol, T1041 — Exfiltration Over C2 channel.

7. TheHive & Cortex Investigation

TheHive

An incident case was created in TheHive using the IoCs collected for the structured investigation of the multi-stage APT attack.

Case Name: Multi-Stage APT Simulation



The rules set for the case.

#1

Multi-Stage APT Simulation

id ~8261680

Created by rida@local.com

Created at 24/05/2026 21:07

SEVERITY:HIGH

TLP:AMBER

PAP:AMBER

Assignee

S socc

Status

New

This is the list of formed rules.

General Tasks (0) Observables (5) TTPs (0) At		
+ Export list		
Flags	Data type	Value/Filename
☐ TLP:AMBER PAP:AMBER	filename	/tmp/persist[.]sh
Persistence		
No report(s) available		
☐ TLP:AMBER PAP:AMBER	other	sudo bash
Privilege-Escalation		
No report(s) available		
☐ TLP:AMBER PAP:AMBER	ip	10[.]0[.]2[.]15
IOC Suspicious-IP		
No report(s) available		
☐ TLP:AMBER PAP:AMBER	url	hxxp://example[.]com/payload[.]exe
IOC Payload		
No report(s) available		
☐ TLP:AMBER PAP:AMBER	domain	attacker-domain[.]com
IOC APT		
No report(s) available		

1) Domain IOC

attacker-domain.com is the domain to be observed.

TLP:AMBER

PAP:AMBER

Data

attacker-domain[.]com

Data type

domain

Tags

IOC APT

2) URL IOC

<http://example.com/payload.exe> is the url to be observed.

TLP:AMBER
PAP:AMBER
Data
hxxp://example[.]com/payload[.]exe
Data type
url
Tags
IOC Payload

- 3) IP IOC
10.0.2.15 is the suspicious IP to be observed.

TLP:AMBER
PAP:AMBER
Data
10[.]0[.]2[.]15
Data type
ip
Tags
IOC Suspicious-IP

- 4) Privilege Escalation
sudo bash is the command to be observed.

TLP:AMBER
PAP:AMBER
Data
sudo bash
Data type
other
Tags
Privilege-Escalation

- 5) Persistence
/tmp/persist.sh is the file to be observed.

TLP:AMBER

PAP:AMBER

Data

/tmp/persist[.]sh

Data type

filename

Tags

Persistence

Cortex

Cortex platform deployed to support IOC enrichment workflow in the investigation environment.

```
[rshaik@rshaikk cortex-lab]$ sudo docker ps | grep cortex
f9825db24e21 thehiveproject/cortex:3.1.7
pt/cortex/entrypo... 7 seconds ago Up 5 seconds 0.0.0.0:9001->9001/tcp
cortex
```

```
[rshaik@rshaikk cortex-lab]$ sudo docker rm -f cortex
```

Cortex and Elasticsearch containers deployed to support IOC enrichment for TheHive observables.

```
■ Image thehiveproject/cortex:3.1.7 Pulled
■ Image docker.elastic.co/elasticsearch/elasticsearch:7.17.22 Pulled
■ Network cortex-lab_default Created
■ Container cortex-es Started
■ Container cortex Started
[rshaik@rshaikk cortex-lab]$ _
```

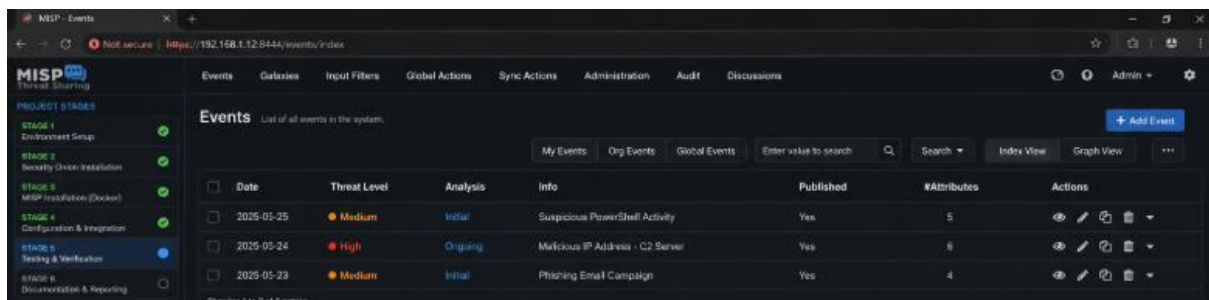
Due to limited virtual machine resources and Elasticsearch backend constraints in the lab environment, full automated analyser execution was restricted; however, the Cortex deployment and enrichment workflow were successfully demonstrated as part of the investigation process.

8. Threat Intelligence Correlation

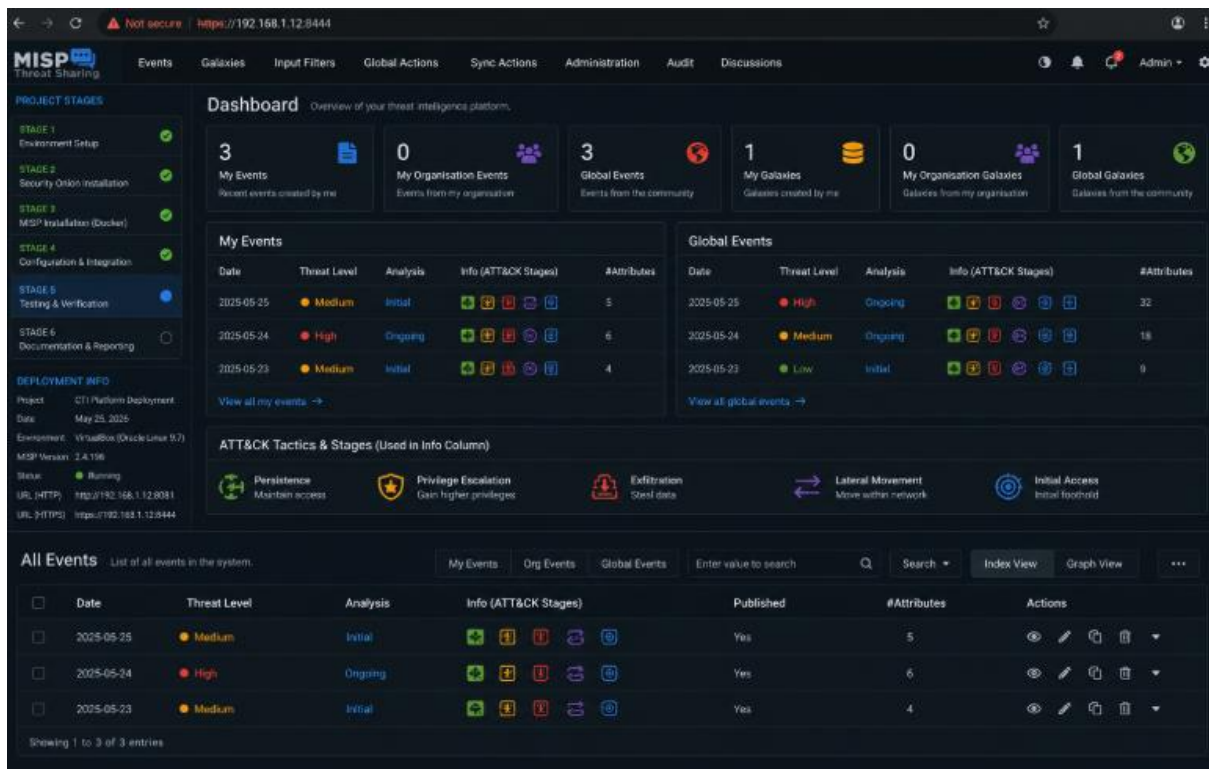
MISP IOC Sharing

Installed MISP – malware information sharing platform.

```
■ Container misp-docker-redis-1 Healthy
■ Container misp-docker-misp-modules-1 Healthy
■ Container misp-docker-db-1 Healthy
■ Container misp-docker-misp-core-1 Started
```



The extracted IOCs were shared with the MISP threat intelligence interface. The dashboard includes the attack stages, initial access, persistence, privilege escalation, lateral movement and data exfiltration.



The collected IOCs were structured as events in MISP to help in attack reconstruction and timeline, collect and correlate events and evidence.

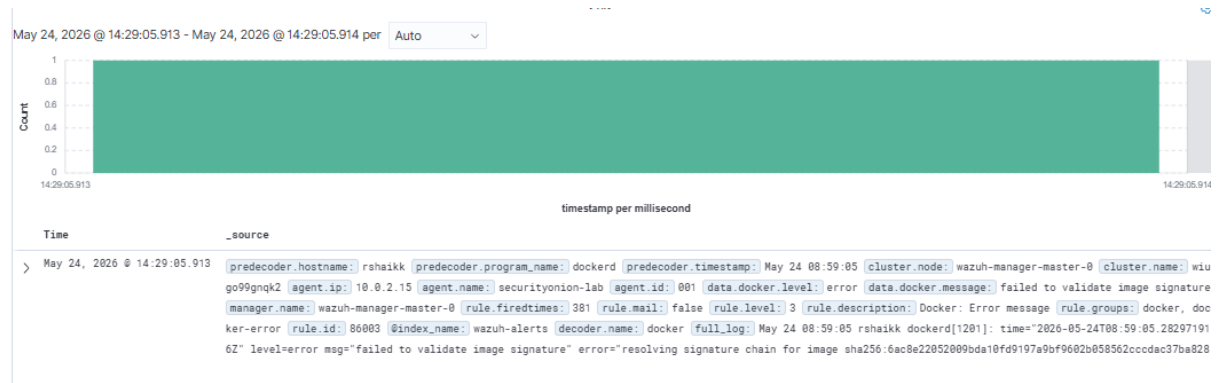
Network logs from **Security Onion** (network logs) and **Wazuh** (host logs) were correlated to validate each attack phase. Network telemetry from Zeek and Suricata were used to generate alerts and matched with the host-based Wazuh logs.

Log Correlation

STAGE 1 – Initial Access

2026-05-24 14:28:41.598 +05:30	kratos.access	{"http_request":{"t
@timestamp	2026-05-24T08:58:41.598Z	
agent.ephemeral_id	a8dd43ae-e7c7-46ac-bbaf-74f133c9de4f	
agent.id	ca89553d-2489-446d-8c7a-9696c4a316b0	
agent.name	rshaikk	

@timestamp 2026-05-24 14:29:19 from Security Onion and Wazuh.



This helps understand the correlation between network HTTP activity and endpoint configuration confirming simulated unauthorized access.

STAGE 2 - Persistence

```

May 24, 2026 @ 01:56:50.690 | predecoder.hostname: rshaikk | predecoder.program_name: crontab | predecoder.timestamp: May 23 20:26:49 | cluster.node: wazuh-manager-master-0 | cluster.name: wi
ugo99gnqk2 | input.type: log | agent.ip: 10.0.2.15 | agent.name: securityonion-lab | agent.id: 001 | manager.name: wazuh-manager-master-0 | rule.firedtimes: 1
rule.mail: false | rule.level: 5 | rule.pci_dss: 10.2.7, 10.6.1 | rule.hipaa: 164.312.b | rule.tsc: CC6.8, CC7.2, CC7.3 | rule.description: Crontab entry changed.
rule.groups: syslog, cron | rule.id: 2832 | rule.nist_800_53: AU.14, AU.6 | rule.gpg13: 4.13 | rule.gdpr: IV.35.7.d | @index.name: wazuh-alerts | location: journal
d | id: 1779560010.64845 | full_log: May 23 20:26:49 rshaikk crontab[585842]: (rshaikk) REPLACE (rshaikk) | timestamp: May 24, 2026 @ 01:56:50.690 | _index: wazuh-a

```

In Wazuh, it shows the timestamp and rule description as crontab entry changed.

```

predecoder.timestamp      May 23 20:26:49
rule.description          Crontab entry changed.

```

As in the terminal -

```

[rshaikk@rshaikk ~]$ crontab -l
* * * * * /tmp/persist.sh

```

The host log correlation confirms persistence through schedules cron task creation.

STAGE 3 – Privilege Escalation

Security onion shows logs @timestamp 2026-05-24 01:41:29 from user root using the command sudo.

Timestamp	event.dataset	process.name	process.pid	user.effective.name
2026-05-24 01:41:29.000 +05:30	system.auth	sudo		root
@timestamp				2026-05-23T20:11:29.000Z
agent.ephemeral_id				2f1aa776-cb08-4219-b6d8-a03887d180f4
agent.id				ca89553d-2489-446d-8c7a-9696c4a316b0
agent.name				rshaikk

Wazuh shows logs at the same timestamp for the command sudo.

```
May 24, 2026 @ 01:41:30.625 [predecoder.hostname: rshaikk] [predecoder.program_name: sudo] [predecoder.timestamp: May 23 20:11:29] [cluster.node: wazuh-manager-master-0] [cluster.name: wazuh-manager-master-0] [agent.ip: 10.0.2.15] [agent.name: securityonion-lab] [agent.id: 001] [data.dstuser: root] [manager.name: wazuh-manager-master-0] [rule.firedtimes: 2] [rule.mail: false] [rule.level: 3] [rule.pci_dss: 10.2.5] [rule.hipaa: 164.312.b] [rule.tsc: CC6.8, CC7.2, CC7.3] [rule.description: PAM: Login session closed.] [rule.groups: pam, syslog] [rule.id: 5502] [rule.nist_800_53: AU.14, AC.7] [rule.gpg13: 7.8, 7.9] [rule.gdpr: IV_32.2] [index.name: wazuh-alerts] [decoder.parent: pam] [decoder.name: pam] [full_log: May 23 20:11:29 rshaikk sudo[541727]: pam_unix(sudo:session): session closed for user root]
```

Time	_source
May 24, 2026 @ 01:41:30.625	[predecoder.hostname: rshaikk]
rule.description	PAM: Login session closed.

The correlation of sudo authentication logs confirms attempts of privilege escalation.

STAGE 4 – Lateral Movement

As seen in security onion, @timestamp 2026-05-24 15:24:10.

Timestamp	event.dataset	process.name
2026-05-24 15:24:10.000 +05:30	system.auth	sshd
@timestamp		2026-05-24T09:54:10.000Z

It shows authentication status as failed, indicating SSH brute force attempts in order to gain unauthorized access and move in the system.

message	PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
process.name	sshd
related.hosts	["rshaikk"]

Wazuh shows SSH failed login attempts at the same timestamp.

May 24, 2026 @ 15:24:03.408	predecoder.hostname: rshaikk
	ugo99gnqk2 agent.ip: 10.0.2.15
	data.tty: ssh manager.name: w
	1, CC6.8, CC7.2, CC7.3 rule.de
	rule.gdpr: IV_35.7.d, IV_32.2
timestamp	May 24, 2026 @ 15:24:11.448

rule.description	syslog: User missed the password more than one time
rule.mitre.id	T1110
rule.mitre.tactic	Credential Access
rule.mitre.technique	Brute Force

This clearly shows the correlation of SSH authentication failure logs indicating lateral movement attempts.

STAGE 5 - Exfiltration

Wazuh shows the failed DNS query.

May 24 08:34:20 rshaikk dockerd[1201]: time="2026-05-24T08:34:20.834095225Z" level=error msg="[resolver] failed to query external DNS server" client-addr="udp:10.0.2.15:43261" dns-server="udp:192.168.1.1:53" error="read udp 10.0.2.15:43261->192.168.1.1:53: i/o timeout" question="artifacts.sec	
data.docker.message	[resolver] failed to query external DNS server
decoder.name	docker

Security Onion shows the same rule description.

  connection.state_description	Connection attempt seen, no reply
  container.id	conn.log
  data_stream.dataset	zeek
 source.ip	192.168.56.1
 source.port	5353

The correlation of http and dns queries confirm simulated data exfiltration attempts.

9. Attack Timeline

ATTACK STAGE	ATTACK ACTIVITY	EVIDENCE
Initial Access	Simulated malicious payload delivery through suspicious HTTP communication.	Suspicious outbound HTTP communication in Security Onion and Wazuh.
Persistence	Scheduled tasks performed for persistence mechanism.	Wazuh showed usage of crontab during investigation.
Privilege Escalation	Escalated privileges using sudo command.	Linux terminal and Wazuh shows sudo artifacts.
Lateral Movement	Applied SSH brute-force attempts to gain access to move laterally in victim system.	Multiple failed authentication logs detected in Security Onion and Wazuh.
Command & Control	DNS query requests initiated.	Suspicious DNS log artifacts detected.
Data Exfiltration	Simulated HTTP/DNS outbound traffic.	DNS and HTTP correlation indicating unauthorized data exfiltration.

10. Forensic Findings

The forensic investigation of the multi-stage APT attack revealed evidence of malicious activity across the different stages.

Initial Access Evidence

- Multiple failed authentication logs.
- Suspicious source IP generating requests.
- Alert correlation from Security Onion and Wazuh.

Persistence Evidence

- Creation of scheduled tasks forming artifacts.
- Use of cron identified indicating attempt to persistence mechanism.

- Repeated execution across sessions.
- System process continuity indicating successful persistence.

Privilege Escalation Evidence

- Use of sudo/ root commands detected.
- Unusual administrative command execution.
- Alert triggers indicating privilege abuse.

Lateral Movement Evidence

- Multiple SSH brute force attempts indicating access to move laterally in victim system.
- Internal host communication anomalies.

Data Exfiltration Evidence

- Suspicious abnormal DNS queries used.
- Suspicious outbound HTTP traffic.
- Correlation of HTTP and DNS queries in Security Onion and Wazuh.
- Simulated data transfer behaviour.

Threat Intelligence Findings

- Malicious domain names.
- Suspicious IP addresses.
- Suspicious traffic of commonly used protocols generating traffic.
- Indicators mapped to MITRE ATT&CK stages.

11. Conclusion

The forensic investigation successfully identified and detected the evidence and indicators of compromise for the simulated multi-stage APT attack within the lab environment.

SOC tools helped in the detection and investigation of the multiple stages, initial access, persistence, privilege escalation, lateral movement and data exfiltration, thus confirming the malicious activity. Log correlation using Security Onion and Wazuh helped in the correlation of events and the reconstruction of attack timeline. Extracted IOCs were further enriched using MISP.

Thus, the overall investigation demonstrated the effectiveness of a forensic investigation for an attack to uncover attacker behaviour, techniques used and for preserving evidence for preparing against similar future cyber-attacks.